WARDA SHAHZAD

PhD Materials Chemistry • Kesterite (CZTS) Thin-Film Photovoltaics • Defect Engineering & Structural-Optical Characterization

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RESEARCH PROFILE

PhD researcher in Materials Chemistry with a focused, publication-backed research line in Cu_2ZnSnS_4 (CZTS) kesterite thin-film solar absorbers. Specializes in solution-based synthesis and alkali-cation (Na, Li, K) doping for defect engineering, secondary-phase suppression, and optical bandgap tuning — the core levers used across the field to push kesterite devices toward higher efficiency. Skilled at building tight composition–processing–property narratives using XRD, SEM/EDX, and UV-Vis spectroscopy, and at translating that work into first-author publications in ranked international journals. Seeking a postdoctoral position to extend this expertise to new synthesis platforms and semiconductor material systems.

KEY EXPERTISE

- **Materials System:** CZTS kesterite thin-film absorbers — alkali (Na, Li, K) doping, defect chemistry, secondary-phase (ZnS, Cu_2SnS_3) suppression, optical bandgap tuning
- **Synthesis:** Solution-based / chemical deposition routes for functional thin films — a cost-effective, scalable approach that broadens compositional screening ahead of vacuum-based deposition
- **Characterization:** X-ray Diffraction (XRD), SEM/EDX, UV-Vis spectroscopy, and structure–property correlation across composition and processing conditions
- **Scientific Writing:** 3 first-author peer-reviewed articles, 1 peer review assignment (Springer Nature), and independently authored research proposals for competitive fellowship applications
- **Research Impact:** ~120 citations, h-index 5, i10-index 5 (Google Scholar)

EDUCATION & TRAINING

PhD, Chemistry — COMSATS University Islamabad, Abbottabad Campus, Pakistan

Fall 2020 – Spring 2026 | Thesis submitted for foreign evaluation; external viva pending

- Thesis: "Exploration of the Physical Properties of Kesterite Thin Films Modified by Alkali Metal Cations for Sunlight Harvesting."
- Doped Cu_2ZnSnS_4 (CZTS) kesterite thin films with Na, Li, and K to engineer defects, suppress secondary phases, and tune optical properties for photovoltaic absorbers; work underpins **two** published first-author articles (Journal of Physics and Chemistry of Solids, 2025; Optical Materials, 2025) and a third under review.
- Milestones: comprehensive examination (Fall 2021), synopsis defense (Fall 2022), pre-DAC (Fall 2025), DAC and internal seminar (Spring 2026).
- Supervisor: Prof. Dr. Bushra Ismail | Co-Supervisor: Prof. Dr. Asad Muhammad Khan

MS, Chemistry — COMSATS University Islamabad, Abbottabad Campus, Pakistan

Fall 2017 – Spring 2019

- Thesis: "Visible Light Driven Photocatalysis based on Alkaline Earth Metal Zirconate Perovskites/Graphene Oxide Composite for Wastewater Treatment"; research contributed to a publication in Ceramics International (2022).
- Supervisor: Prof. Dr. Bushra Ismail | Co-Supervisor: Prof. Dr. Rafaqat Ali Khan

MSc, Chemistry — University of Wah, Pakistan

Fall 2015 – Spring 2017

- 4th-semester research internship, Pakistan Ordnance Factories (POF) Institute of Technology, SJL Dept, Wah Cantt (1 July – 1 August 2017).

RESEARCH EXPERIENCE

Doctoral Researcher — CZTS Thin-Film Solar Absorbers

Department of Chemistry, COMSATS University Islamabad, Abbottabad Campus | 2020 – 2026

- Designed and optimized solution-based synthesis routes for Na-, Li-, and K-doped CZTS kesterite thin films, generating a systematic, reproducible dataset across multiple alkali dopant systems.

- Resolved how alkali-doping level governs defect chemistry, secondary-phase formation (ZnS, Cu_2SnS_3), and optical bandgap — building a mechanistic picture directly relevant to absorber-layer efficiency losses.
- Independently ran XRD, SEM/EDX, and UV-Vis spectroscopy workflows, from data collection through structural, morphological, and optical interpretation.
- Converted findings into two first-author journal articles and a third manuscript currently under review, while managing the full PhD milestone sequence (comprehensive exam, synopsis, DAC, thesis) on schedule.
- **Technique scope:** solution-based/chemical synthesis rather than physical vapor deposition (co-evaporation, sputtering, e-beam, thermal evaporation) — a complementary, cost-effective route for compositional screening ahead of vacuum-based scale-up.

PUBLICATIONS & RESEARCH OUTPUT

~120 citations | h-index 5 | i10-index 5 *(Google Scholar, self-reported)*

First-Author — PhD Research

- Journal of Physics and Chemistry of Solids (2025) — doi.org/10.1016/j.jpcs.2025.113116
- Optical Materials (2025) — doi.org/10.1016/j.optmat.2025.117582
- Third first-author article (CZTS kesterite) — submitted, currently under review

First-Author — MS Research

- Ceramics International (2022) — doi.org/10.1016/j.ceramint.2022.05.151

Co-Authored

- Materials Science and Engineering B — doi.org/10.1016/j.mseb.2024.117273
- Journal of Inorganic and Organometallic Polymers — link.springer.com/article/10.1007/s10904-025-03621-x
- Additional co-authored publications in RSC Advances and related venues.

Peer Review Service

- Reviewer, Journal of Materials Science: Materials in Electronics (Springer Nature), 2025.

CONFERENCES, AWARDS & TRAINING

- **2nd Prize, Poster Presentation** — SSCS Conference, COMSATS University Islamabad, Abbottabad Campus, April 2025
- **Oral Presentation** — ICRA-C 2024, NUST Islamabad, August 2024
- **Poster Presentation** — ICC RTC&RET, University of Swat, June 2023
- Hands-on Training Workshop in Advanced Analytical Instrumentation, National Centre for Physics, Islamabad, 29 June – 3 July 2026 (Participant)
- International Conference "Nanomaterials: New Trends in Development and Applications," Forman Christian College University, Lahore, 29–31 January 2019 (Participant)
- International Workshop on Energy Materials and Nanotechnology (EMNT 2018), COMSATS University Islamabad, Abbottabad Campus, 15–16 August 2018 (Participant)

TECHNICAL SKILLS

- **Synthesis:** Solution-based / chemical deposition routes for kesterite (CZTS) thin-film absorbers
- **Characterization:** X-ray Diffraction (XRD); Scanning Electron Microscopy / Energy-Dispersive X-ray Spectroscopy (SEM/EDX); UV-Vis Spectroscopy; advanced analytical instrumentation (NCP Islamabad, 2026)
- **Materials Systems:** Kesterite/chalcogenide defect engineering, alkali-cation doping, secondary-phase analysis, photocatalytic and energy-related functional materials
- **Scientific Communication:** Manuscript writing and peer review; research proposal development for competitive fellowship applications

LANGUAGES

- Urdu — Native
- English — Fluent (professional / academic working proficiency)

REFERENCES

Prof. Dr. Bushra Ismail — *PhD Supervisor*
Professor, Department of Chemistry, COMSATS University Islamabad, Abbottabad Campus, Pakistan | bushraismail@cuiatd.edu.pk

Prof. Dr. Asad Muhammad Khan — *PhD Co-Supervisor*
Professor, Department of Chemistry, COMSATS University Islamabad, Abbottabad Campus, Pakistan | amk@cuiatd.edu.pk

Prof. Dr. Rafaqat Ali Khan — *Supervisory Committee Member*
Professor, Department of Chemistry, COMSATS University Islamabad, Abbottabad Campus, Pakistan | rafaqatalikhan@cuiatd.edu.pk